

LAKES / INDIA LAKES AND WETLANDS

CATEGORY BOUNDARY → LAKE-ORIGIN TREE → WATER-BUDGET EQUATION → LAKE-CHANGE FIREWALL → WETLAND HYDROPERIOD → INDIA LAKE MAP

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

PRIMARY CORE FLOW SUBORDINATE EXTRA APPROVAL / REVIEW

Approval: FALSE • Pending user review • active generation g2 • explicit approval of this exact generation is required • all prior artifacts unchanged

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles

00

STAGE 00

CATEGORY BOUNDARY

CATEGORY BOUNDARYSTANDING WATER IN AN INLAND BASINARTIFICIAL OR STRONGLY REGULATED STORAGEBROADER HYDROLOGICAL-ECOLOGICAL CATEGORYRAMSAR SITEINTERNATIONAL DESIGNATION, NOT BASIN ORIGIN

DEFINITION / COMPONENTS	TRIGGER / MECHANISM	EFFECT / INTERVENTION
LAKE → standing water in an inland basin	WETLAND → broader hydrological-ecological category	LAKE → standing water in an inland basin
RESERVOIR → artificial or strongly regulated storage	RAMSAR SITE → international designation, not basin origin	RESERVOIR → artificial or strongly regulated storage

DEFINITION / COMPONENTS

• LAKE → standing water in an inland basin

• RESERVOIR → artificial or strongly regulated storage

TRIGGER / MECHANISM

• WETLAND → broader hydrological-ecological category

• RAMSAR SITE → international designation, not basin origin

EFFECT / INTERVENTION

• LAKE → standing water in an inland basin

• RESERVOIR → artificial or strongly regulated storage

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RAMSAR SITE → international designation, not basin origin.

01

STAGE 01

LAKE-ORIGIN TREE

LAKE-ORIGIN TREEEARTH MOVEMENTTECTONIC OR VOLCANIC BASINTARN, OX-BOW OR LAGOONLANDSLIDE DAM OR CRATERHUMAN WORKRESERVOIR, BUND OR EXCAVATION

ICE / RIVER / COAST → tarn, ox-bow or lagoon

BLOCKAGE / IMPACT → landslide dam or crater

HUMAN WORK → reservoir, bund or excavation

EARTH MOVEMENT → tectonic or volcanic basin

DEFINITION / COMPONENTS	TRIGGER / MECHANISM	EFFECT / INTERVENTION
• EARTH MOVEMENT → tectonic or volcanic basin	• BLOCKAGE / IMPACT → landslide dam or crater	• EARTH MOVEMENT → tectonic or volcanic basin
• ICE / RIVER / COAST → tarn, ox-bow or lagoon	• HUMAN WORK → reservoir, bund or excavation	• ICE / RIVER / COAST → tarn, ox-bow or lagoon

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: BLOCKAGE / IMPACT → landslide dam or crater.

02

STAGE 02

WATER-BUDGET EQUATION

WATER-BUDGET EQUATIONRAIN + RIVER + GROUNDWATEREVAPORATION + OUTFLOW + SEEPAGESTORAGE CHANGINPUTS MINUS OUTPUTSSTATE PERIOD, BASIN AND UNCERTAINTY

INPUT → rain + river + groundwater

OUTPUT → evaporation + outflow + seepage

STORAGE CHANGE → inputs minus outputs

ATTRIBUTION → state period, basin and uncertainty

DEFINITION / COMPONENTS	TRIGGER / MECHANISM	EFFECT / INTERVENTION
• INPUT → rain + river + groundwater	• STORAGE CHANGE → inputs minus outputs	• INPUT → rain + river + groundwater
• OUTPUT → evaporation + outflow + seepage	• ATTRIBUTION → state period, basin and uncertainty	• OUTPUT → evaporation + outflow + seepage

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: STORAGE CHANGE → inputs minus outputs; ATTRIBUTION → state period, basin and uncertainty.

03

STAGE 03

LAKE-CHANGE FIREWALL

LAKE-CHANGE FIREWALLSEDIMENT AND ORGANIC INFILLINGNUTRIENT-PRODUCTIVITY-OXYGEN SHIFTWATER-BUDGET DEFICITSIMILAR APPEARANCE CAN HIDE DIFFERENT MECHANISMS

DEFINITION / COMPONENTS	TRIGGER / MECHANISM	EFFECT / INTERVENTION
SUCCESSION → sediment and organic infilling	SHRINKAGE → water-budget deficit	SUCCESSION → sediment and organic infilling
EUTROPHICATION → nutrient-productivity-oxygen shift	RULE → similar appearance can hide different mechanisms	EUTROPHICATION → nutrient-productivity-oxygen shift

DEFINITION / COMPONENTS

TRIGGER / MECHANISM

EFFECT / INTERVENTION

03

STAGE 03

LAKE-CHANGE FIREWALL

LAKE-CHANGE FIREWALLSEDIMENT AND ORGANIC INFILLINGNUTRIENT-PRODUCTIVITY-OXYGEN SHIFTWATER-BUDGET DEFICITSIMILAR APPEARANCE CAN HIDE DIFFERENT MECHANISMS

DEFINITION / COMPONENTS	TRIGGER / MECHANISM	EFFECT / INTERVENTION
SUCCESSION → sediment and organic infilling	SHRINKAGE → water-budget deficit	SUCCESSION → sediment and organic infilling
EUTROPHICATION → nutrient-productivity-oxygen shift	RULE → similar appearance can hide different mechanisms	EUTROPHICATION → nutrient-productivity-oxygen shift

DEFINITION / COMPONENTS

- SUCCESSION → sediment and organic infilling
- EUTROPHICATION → nutrient-productivity-oxygen shift

TRIGGER / MECHANISM

- SHRINKAGE → water-budget deficit
- RULE → similar appearance can hide different mechanisms

EFFECT / INTERVENTION

- SUCCESSION → sediment and organic infilling
- EUTROPHICATION → nutrient-productivity-oxygen shift

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: STORAGE CHANGE → inputs minus outputs; ATTRIBUTION → state period, basin and uncertainty.

04

STAGE 04

WETLAND HYDROPERIOD

WETLAND HYDROPERIODSEASON OF INUNDATIONHOW LONG WATER REMAINSDEPTH + FREQUENCYHABITAT FILTERRIVER, AQUIFER OR COAST LINK

TIMING → season of inundation

DURATION → how long water remains

DEPTH + FREQUENCY → habitat filter

CONNECTIVITY → river, aquifer or coast link

DEFINITION / COMPONENTS	TRIGGER / MECHANISM	EFFECT / INTERVENTION
TIMING → season of inundation	DEPTH + FREQUENCY → habitat filter	TIMING → season of inundation
DURATION → how long water remains	CONNECTIVITY → river, aquifer or coast link	DURATION → how long water remains

DEFINITION / COMPONENTS

- TIMING → season of inundation
- DURATION → how long water remains

TRIGGER / MECHANISM

- DEPTH + FREQUENCY → habitat filter
- CONNECTIVITY → river, aquifer or coast link

EFFECT / INTERVENTION

- TIMING → season of inundation
- DURATION → how long water remains

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: DEPTH + FREQUENCY → habitat filter; CONNECTIVITY → river, aquifer or coast link.

05

STAGE 05

INDIA LAKE MAP

INDIA LAKE MAPJHELUM-LINKED KASHMIR BASINSALINE RAJASTHAN CLOSED BASINODISHA COASTAL LAGOON

SAMBHAR → saline Rajasthan closed basin

CHILIKA → Odisha coastal lagoon

WULAR → Jhelum-linked Kashmir basin

LONAR / KANWAR / BHOJ → impact / fluvial / artificial

SYSTEM / DEFINITION	PROCESS / MECHANISM	SPATIAL / INDIA PATTERN	HAZARD / LIMIT / RESPONSE
- WULAR → Jhelum-linked Kashmir basin - SAMBHAR → saline Rajasthan closed basin	- CHILIKA → Odisha coastal lagoon - LONAR / KANWAR / BHOJ → impact / fluvial / artificial	- WULAR → Jhelum-linked Kashmir basin - SAMBHAR → saline Rajasthan closed basin	- CHILIKA → Odisha coastal lagoon - LONAR / KANWAR / BHOJ → impact / fluvial / artificial

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: LONAR / KANWAR / BHOJ → impact / fluvial / artificial.

06

STAGE 06

RIVER-LAKE TRAPS

RIVER-LAKE TRAPSLINKED TO JHELUMBETWEEN KRISHNA AND GODAVARI DELTASPHUMDI-BEARING LAKEADJACENCY DOES NOT PROVE DIRECT FEEDING

DEFINITION / COMPONENTS	TRIGGER / MECHANISM	EFFECT / INTERVENTION
WULAR → linked to Jhelum	LOKTAK → phumdi-bearing lake	WULAR → linked to Jhelum
KOLLERU → between Krishna and Godavari deltas	RULE → adjacency does not prove direct feeding	KOLLERU → between Krishna and Godavari deltas

DEFINITION / COMPONENTS

- WULAR → linked to Jhelum
- KOLLERU → between Krishna and Godavari deltas

TRIGGER / MECHANISM

- LOKTAK → phumdi-bearing lake
- RULE → adjacency does not prove direct feeding

EFFECT / INTERVENTION

- WULAR → linked to Jhelum
- KOLLERU → between Krishna and Godavari deltas

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: LONAR / KANWAR / BHOJ → impact / fluvial / artificial.

06

STAGE 06 RIVER-LAKE TRAPS

RIVER-LAKE TRAPS LINKED TO JHELUM BETWEEN KRISHNA AND GODAVARI DELTAS PHUMDI-BEARING LAKE ADJACENCY DOES NOT PROVE DIRECT FEEDING

DEFINITION / COMPONENTS	TRIGGER / MECHANISM	EFFECT / INTERVENTION
WULAR → linked to Jhelum	LOKTAK → phumdi-bearing lake	WULAR → linked to Jhelum
KOLLERU → between Krishna and Godavari deltas	RULE → adjacency does not prove direct feeding	KOLLERU → between Krishna and Godavari deltas
DEFINITION / COMPONENTS - WULAR → linked to Jhelum - KOLLERU → between Krishna and Godavari deltas	TRIGGER / MECHANISM - LOKTAK → phumdi-bearing lake - RULE → adjacency does not prove direct feeding	EFFECT / INTERVENTION - WULAR → linked to Jhelum - KOLLERU → between Krishna and Godavari deltas

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → adjacency does not prove direct feeding.

07

STAGE 07 URBAN RECLAMATION CHAIN

URBAN RECLAMATION CHAIN STORAGE AND DRAINAGE LOST FLOOD PEAK AND DURATION RISE DILUTION AND HABITAT DECLINE LOW-LYING COMMUNITIES BEAR SHIFTED RISK

FILL / NARROW → storage and drainage lost	→	RUNOFF → flood peak and duration rise	→	POLLUTION → dilution and habitat decline	→	EQUITY → low-lying communities bear shifted risk
DEFINITION / COMPONENTS - FILL / NARROW → storage and drainage lost - RUNOFF → flood peak and duration rise		TRIGGER / MECHANISM - POLLUTION → dilution and habitat decline - EQUITY → low-lying communities bear shifted risk		EFFECT / INTERVENTION - FILL / NARROW → storage and drainage lost - RUNOFF → flood peak and duration rise		

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: FILL / NARROW → storage and drainage lost.

08

STAGE 08 THREAT-RESPONSE MATRIX

THREAT-RESPONSE MATRIX SOURCE CONTROL + TREATMENT CATCHMENT AND EROSION MANAGEMENT PROTECT HYDROLOGICAL SPACE ALTERED FLOW ECOLOGICAL AND FLOW RESTORATION

CLASSIFICATION	DIAGNOSTIC BASIS	PROCESS LINK	EXAM TRAP
SEWAGE / NUTRIENT → source control + treatment	ENCROACHMENT → protect hydrological space	SEWAGE / NUTRIENT → source control + treatment	ENCROACHMENT → protect hydrological space
SEDIMENT → catchment and erosion management	INVASIVE / ALTERED FLOW → ecological and flow restoration	SEDIMENT → catchment and erosion management	INVASIVE / ALTERED FLOW → ecological and flow restoration
CLASSIFICATION - SEWAGE / NUTRIENT → source control + treatment - SEDIMENT → catchment and erosion management	DIAGNOSTIC BASIS - ENCROACHMENT → protect hydrological space - INVASIVE / ALTERED FLOW → ecological and flow restoration	PROCESS LINK - SEWAGE / NUTRIENT → source control + treatment - SEDIMENT → catchment and erosion management	EXAM TRAP - ENCROACHMENT → protect hydrological space - INVASIVE / ALTERED FLOW → ecological and flow restoration

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: INVASIVE / ALTERED FLOW → ecological and flow restoration.

09

STAGE 09 RAMSAR STATUS FIREWALL

RAMSAR STATUS FIREWALL AT LEAST ONE CAN SUPPORT LISTING WISE USE MAINTAIN ECOLOGICAL CHARACTER PRIORITY ATTENTION TO CHARACTER CHANGE LISTING IS NOT AUTOMATIC NATIONAL PARK STATUS

DEFINITION / COMPONENTS	TRIGGER / MECHANISM	EFFECT / INTERVENTION
CRITERION → at least one can support listing	MONTREUX → priority attention to character change	CRITERION → at least one can support listing
WISE USE → maintain ecological character	LIMIT → listing is not automatic National Park status	WISE USE → maintain ecological character
DEFINITION / COMPONENTS - CRITERION → at least one can support listing - WISE USE → maintain ecological character	TRIGGER / MECHANISM - MONTREUX → priority attention to character change - LIMIT → listing is not automatic National Park status	EFFECT / INTERVENTION - CRITERION → at least one can support listing - WISE USE → maintain ecological character

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: LIMIT → listing is not automatic National Park status.

09

STAGE 09 RAMSAR STATUS FIREWALL

RAMSAR STATUS FIREWALL AT LEAST ONE CAN SUPPORT LISTING WISE USE MAINTAIN ECOLOGICAL CHARACTER PRIORITY ATTENTION TO CHARACTER CHANGE LISTING IS NOT AUTOMATIC NATIONAL PARK STATUS

DEFINITION / COMPONENTS	TRIGGER / MECHANISM	EFFECT / INTERVENTION
CRITERION → at least one can support listing	MONTREUX → priority attention to character change	CRITERION → at least one can support listing
WISE USE → maintain ecological character	LIMIT → listing is not automatic National Park status	WISE USE → maintain ecological character

DEFINITION / COMPONENTS

- CRITERION → at least one can support listing
- WISE USE → maintain ecological character

TRIGGER / MECHANISM

- MONTREUX → priority attention to character change
- LIMIT → listing is not automatic National Park status

EFFECT / INTERVENTION

- CRITERION → at least one can support listing
- WISE USE → maintain ecological character

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: LIMIT → listing is not automatic National Park status.

10

STAGE 10 INDIA GOVERNANCE CHAIN

INDIA GOVERNANCE CHAIN IDENTIFY AND DELINEATE SYSTEM REGULATE ACTIVITIES FINANCE AND IMPLEMENT MANAGEMENT PROVE ECOLOGICAL RESPONSE

INVENTORY → identify and delineate system

NOTIFICATION / RULES → regulate activities

PLAN / NPCA → finance and implement management

MONITOR → prove ecological response

SYSTEM / DEFINITION	PROCESS / MECHANISM	SPATIAL / INDIA PATTERN	HAZARD / LIMIT / RESPONSE
- INVENTORY → identify and delineate system - NOTIFICATION / RULES → regulate activities	- PLAN / NPCA → finance and implement management - MONITOR → prove ecological response	- INVENTORY → identify and delineate system - NOTIFICATION / RULES → regulate activities	- PLAN / NPCA → finance and implement management - MONITOR → prove ecological response

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: PLAN / NPCA → finance and implement management.

11

STAGE 11 LAKE-WETLAND ANSWER SPINE

LAKE-WETLAND ANSWER SPINE BASIN, WETLAND AND STATUS ORIGIN AND HYDROLOGY BUDGET, NUTRIENT OR CONNECTIVITY FAILURE BASIN-SCALE, MONITORED AND STATUS-AWARE

DEFINE → basin, wetland and status

CLASSIFY → origin and hydrology

DIAGNOSE → budget, nutrient or connectivity failure

RESPOND → basin-scale, monitored and status-aware

DEFINITION / COMPONENTS	TRIGGER / MECHANISM	EFFECT / INTERVENTION
- DEFINE → basin, wetland and status - CLASSIFY → origin and hydrology	- DIAGNOSE → budget, nutrient or connectivity failure - RESPOND → basin-scale, monitored and status-aware	- DEFINE → basin, wetland and status - CLASSIFY → origin and hydrology

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: DIAGNOSE → budget, nutrient or connectivity failure; RESPOND → basin-scale, monitored and status-aware.

E

EXTRA SUBORDINATE ENRICHMENT — ONLY AFTER THE COMPLETE CORE

OPTIONAL ADVANCED ENRICHMENT INDIAN GLACIAL LAKES IN THE SOURCE ARE HIMALAYAN GANGABAL, SHESHNAG, KAILASH-KUND, SANASAR. NO PLEISTOCENE GLACIATION IN MAJID'S ACCOUNT; HENCE NO PENINSULAR WULAR (J&K)

TECTONIC FRESHWATER LAKE; AMONG ASIA'S LARGEST FRESHWATER LAKES. CHILIKA AND PULICAT EAST-COAST LAGOONS.

OPTIONAL PROCESS DEPTH	DATA / INSTITUTION LIMIT	CONTEMPORARY PARALLEL
- Indian glacial lakes in the source are Himalayan: Gangabal, Sheshnag, Kailash-Kund, Sanasar. - Peninsula: no Pleistocene glaciation in Majid's account; hence no Peninsular glacial tarn belt. - Wular (J&K) = tectonic freshwater lake; among Asia's largest freshwater lakes. - Chilika and Pulicat = east-coast lagoons.	- Sambhar (Rajasthan) = major inland salt lake. - Sasthamkotta = Kerala's largest freshwater lake. - Sat Tal = Kumaun lake group near Bhimtal; avoid reducing its origin to a single glacial label. - NOT: Peninsular India has glacial tarn lakes → Glacial lakes are Himalayan in the source; Peninsular India shows no Pleistocene glaciation .	- NOT: Sambhar is a freshwater lake → Sambhar is a salt lake of Rajasthan's arid inland drainage. - NOT: Pulicat is a west-coast lagoon → Pulicat is on the east coast (AP/TN). - NOT: Wular is a lagoon → Wular is tectonic/freshwater , not coastal lagoonal. - NOT: All Ramsar lakes have the same origin → Ramsar status is conservation status; origin may be glacial, tectonic, lagoonal, man-made or salt.

OPTIONAL STATUS — This optional depth is unnecessary for a competent core answer; use it only for added nuance after the complete core has been written.